

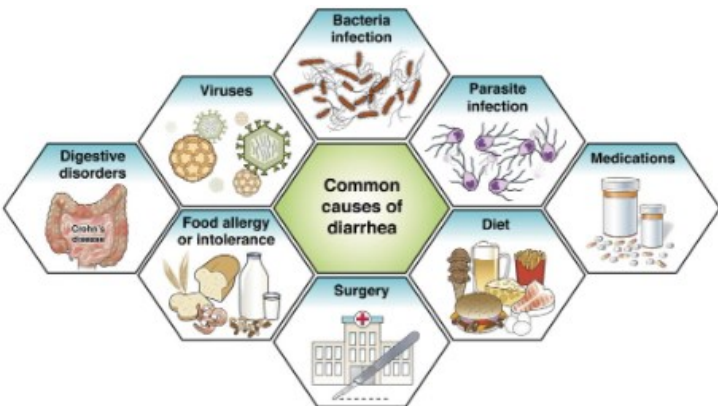


Approach to Adults with Acute Diarrhea

Most cases of acute diarrhea in adults are of infectious etiology, and most cases resolve with symptomatic treatment alone. When clinicians care for adults with diarrhea, two important decision points are (1) when to perform stool testing and (2) whether to initiate empiric antimicrobial therapy.

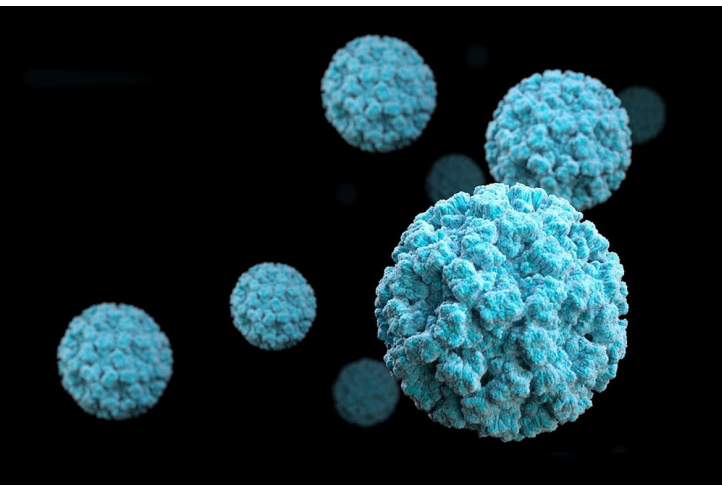
History

Initial evaluation of patients who present to medical care with acute diarrhea should include a careful history to determine the duration of symptoms, the frequency and characteristics of the stool, and associated symptoms. Look for signs of dehydration, too. Diarrhea of **small bowel** origin is typically watery, of large volume, and associated with abdominal cramping, bloating, and gas. Diarrhea of **large intestinal** origin often presents with frequent, regular, small volume, and often painful bowel movements. Fever and bloody or mucoid stools are common, and red blood cells and inflammatory cells can be seen routinely on stool microscopy.



Note. From "What is diarrhea?" reviewed by Rajeev Jain, 2020, in *Clinical Gastroenterology and Hepatology*, 19, xvii.

Details like food hx and timing of onset of symptoms, can provide clues as to the cause of diarrhea.



Note. From *Norovirus virions* [Illustration], by Alissa Eckert, 2016, Centers for Disease Control and Prevention. (<https://www.pennmedicine.org/news/-/media/images/pr%20news/news/2017/october/norovirus.ashx?la=en>).

- **Within six hours:** Suggests ingestion of a preformed toxin of *Staphylococcus aureus* or *Bacillus cereus*, particularly if nausea and vomiting were the initial symptoms.
- **At 8 to 16 hours:** Suggestions infection with *Clostridium perfringens*.
- **At more than 16 hours:** Suggests either viral or other bacterial infection (e.g., contamination of food with enterotoxigenic or STEC or other pathogens).
- Exposure to animals (poultry, turtles, petting zoos) has been associated with *Salmonella* infection.

Travel to a resource-limited setting increases the risk of bacterial diarrhea, and it also informs the risk of certain parasitic infections. **Occupation** in daycare centers has been associated with infections with *Shigella*, *Cryptosporidium*, and *Giardia*. Rotavirus is a potential consideration, too.

Recent antibiotic use should also be taken into consideration.

Physical exam: Look for signs of dehydration, abdominal distention, pain with percussion, rigidity or rebound tenderness.

Lab test/stool testing: Laboratory tests are not routinely warranted for most patients with acute diarrhea. We pursue microbiologic stool testing for patients with acute community-acquired diarrhea and the following features:

Concern	Symptoms
Severe illness	• Hypovolemia • Passage of >6 unformed stools per 24 hours • Severe abdominal pain
Other signs or symptoms concerning for inflammatory diarrhea	• Bloody diarrhea • Passage of many small-volume stools containing blood and mucus • Temperature ≥38.5°C (101.3°F)
High-risk host features	• Age ≥70 years • Comorbidities, such as cardiac disease, which may be exacerbated by hypovolemia or rapid infusion of fluid • Immunocompromising condition (including advanced HIV infection) • Inflammatory bowel disease • Pregnancy

Anyone with bloody diarrhea or persistent diarrhea should be seen and evaluated further.

Management: Rehydration, Brat diet/antimotility agents like Imodium or Pepto Bismol. (Pts with no fever and no bloody stools).

Empiric abx therapy is mainly for severe illness (fever, hypovolemia, severe abdominal pain), bloody diarrhea, or with high-risk host features (see above).

Common abx used are Azithromycin or fluoroquinolone.

Azithromycin can be given as a single 1 g dose (for patients without dysentery) or as a 500 mg once daily for three days (for patients with or without dysentery). Fluoroquinolones include [ciprofloxacin](#) (a single 750 mg dose or 500 mg twice daily for three to five days) or [levofloxacin](#) (500 mg as a single dose or given once daily for three to five days).

If diarrhea resolves or responds rapidly to therapy, no further work-up or treatment is necessary. If diarrhea becomes persistent, the search for an etiology should be expanded to try to isolate a treatable process or pathogen, or identify a noninfectious etiology.

References

Eckert, A. (2016). Norovirus virions. Centers for Disease Control and Prevention. <https://www.pennmedicine.org/news/-/media/images/pr%20news/news/2017/october/norovirus.ashx?la=en>

Jain, R. (2020). What is diarrhea? *Gastroenterology and Hepatology*, 19, xvii. <https://www.cghjournal.org/action/showPdf?pii=S1542-3565%2820%2931387-2>